Activity3 Timeseries R and Tableau

R1. Multiple Time-Series Chart and Linear Trend Analysis

R code:

```
# Generate Dataset
set.seed(40425)
branch <- c("Perambur", "Velachery", "Anna Nagar")
n <- 36
month_no <- 1:36
date <- seq(as.Date("2023-01-01"),
            by = "month", length.out = n)
base <- c(75, 60, 90)
growth <- c(1.2, 0.4, 0.8)
amp <- c(5, 3, 4)
peak_month <- 3
sales <- sapply(1:3, function(i)
  round(base[i] + growth[i] * month_no +
  amp[i] * cos(2*pi*(month_no-peak_month)/12) +
  rnorm(n, 0, 2), 1))
df <- data.frame(
  Month = rep(date, 3),
  Month_no = rep(month_no, 3),
  Branch = rep(branch, each=n),
  Sales = as.vector(sales)
)
```

```
# R1 - Multiple Time Series Chart
per <- subset(df, Branch == "Perambur")
vel <- subset(df, Branch == "Velachery")
anna <- subset(df, Branch == "Anna Nagar")
plot(per$Month, per$Sales,
    type="l", col="blue", lwd=2,
    ylim=range(df$Sales),
    xlab="Month", ylab="Sales (Rs lakh)",
    main="Monthly Sales of Three Branches")
lines(vel$Month, vel$Sales, col="red", lwd=2)
lines(anna$Month, anna$Sales, col="green", lwd=2)
text(tail(per$Month,1), tail(per$Sales,1),
    "Perambur", pos=4, col="blue")
text(tail(vel$Month,1), tail(vel$Sales,1),
    "Velachery", pos=4, col="red")
text(tail(anna$Month,1), tail(anna$Sales,1),
    "Anna Nagar", pos=4, col="green")
# Linear Trend Slopes
m1 <- lm(Sales ~ Month_no, data=per)
m2 <- lm(Sales ~ Month_no, data=vel)
m3 <- lm(Sales ~ Month_no, data=anna)
slopes <- c(coef(m1)[2], coef(m2)[2], coef(m3)[2])
```

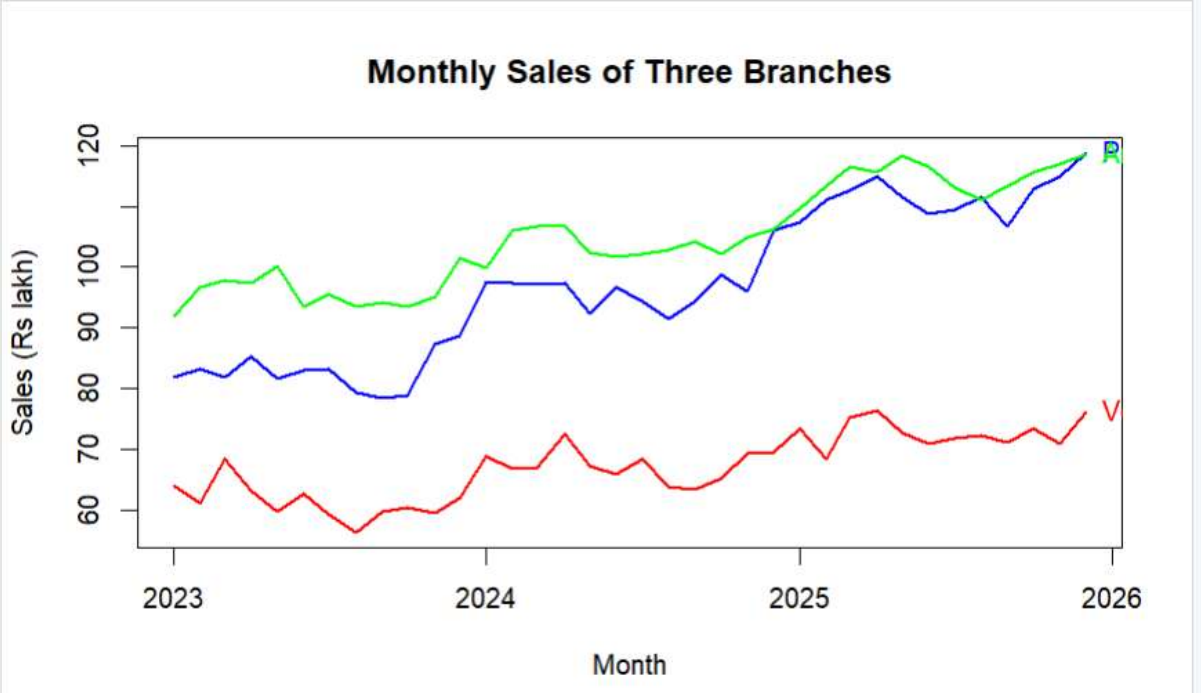
```
names(slopes) <- branch

print(slopes)

cat("Fastest Growing Branch:",

    names(which.max(slopes)))
```

Output:



R2. Seasonal Analysis of Perambur Branch

R code:

```
per_ts <- ts(per$Sales, start=c(2023,1), frequency=12)

fit <- stl(per_ts, s.window="periodic")

plot(fit)

seasonal <- fit$time.series[, "seasonal"]

remainder <- fit$time.series[, "remainder"]

monthly_season <- tapply(seasonal, cycle(per_ts), mean)

peak <- which.max(monthly_season)

trough <- which.min(monthly_season)
```

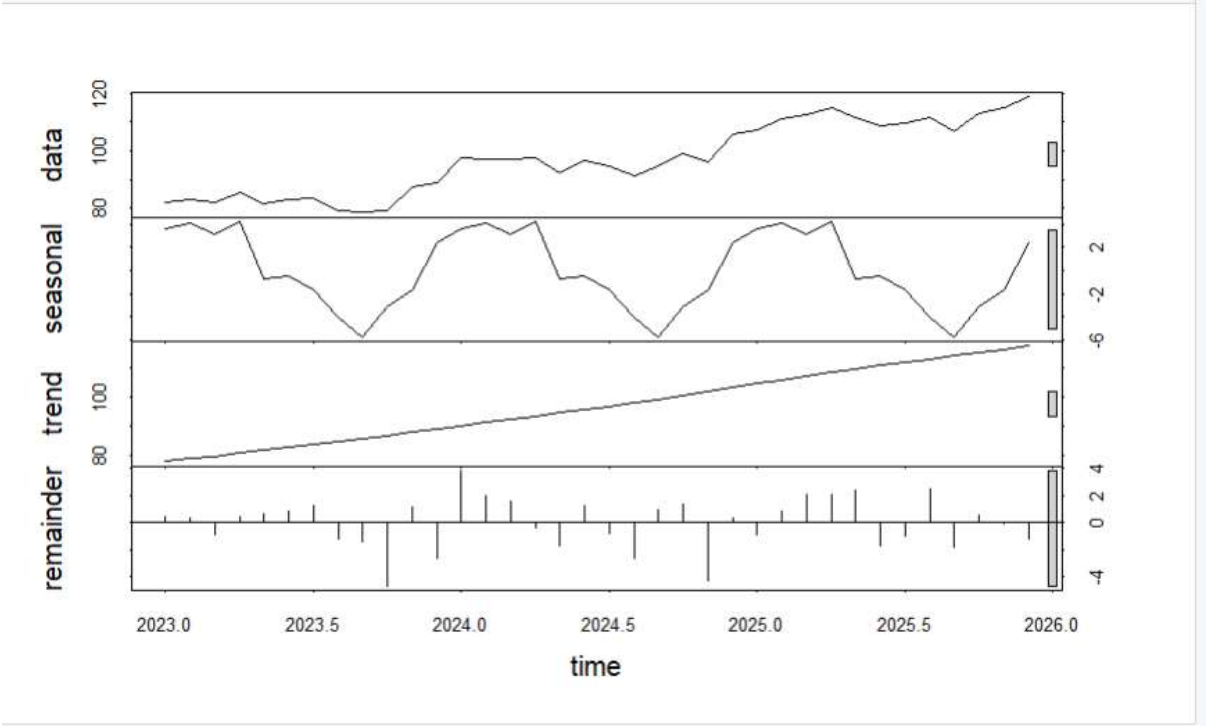
```
strength <- max(0, 1-var(remainder)/var(seasonal+remainder))

cat("Peak Month:", month.name[peak], "\n")

cat("Trough Month:", month.name[trough], "\n")

cat("Seasonal Strength:", strength)
```

Output:



R3. Total Sales and Moving Averages

R code:

```
total_sales <- aggregate(Sales ~ Month_no + Month,

                         data=df, sum)

total_ts <- ts(total_sales$Sales,

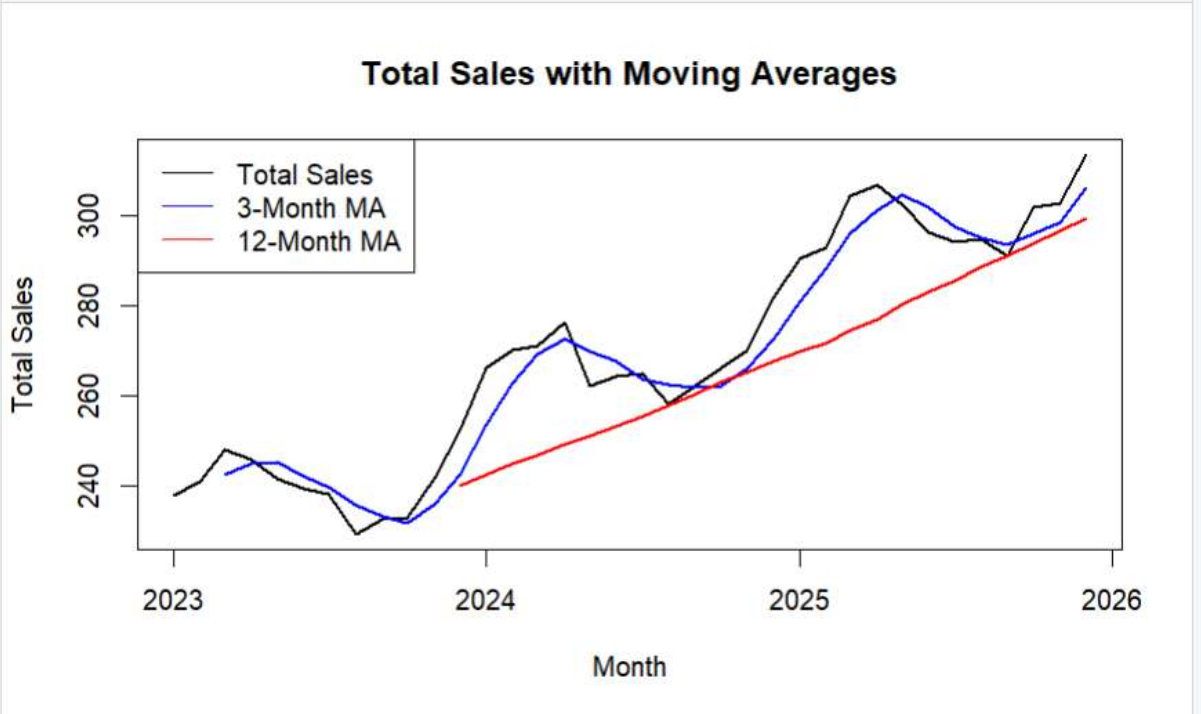
               start=c(2023,1), frequency=12)

ma3 <- stats::filter(total_ts, rep(1/3,3), sides=1)

ma12 <- stats::filter(total_ts, rep(1/12,12), sides=1)
```

```
plot(total_sales$Month, total_sales$Sales,
     type="l", col="black", lwd=2,
     xlab="Month", ylab="Total Sales",
     main="Total Sales with Moving Averages")
lines(total_sales$Month, ma3, col="blue", lwd=2)
lines(total_sales$Month, ma12, col="red", lwd=2)
legend("topleft",
       legend=c("Total Sales","3-Month MA","12-Month MA"),
       col=c("black","blue","red"), lty=1)
cat("3-Month MA at Month 24:", ma3[24], "\n")
cat("12-Month MA at Month 24:", ma12[24], "\n")
```

Output:

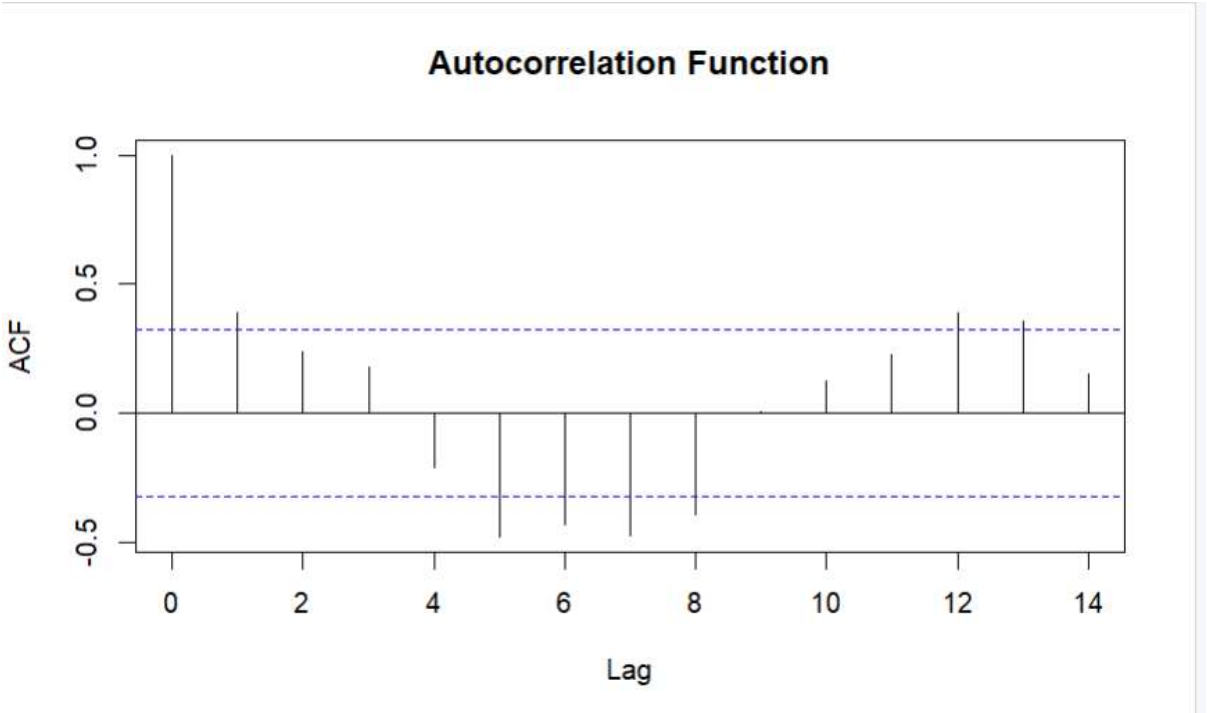


R4. Residual Analysis and Autocorrelation

R code:

```
vel_model <- lm(Sales ~ Month_no, data=vel)
res <- residuals(vel_model)
plot(res, type="l", col="blue",
    main="Velachery Residual Analysis",
    xlab="Month", ylab="Residuals")
abline(h=0, col="red", lty=2)
acf(res, lag.max=14, main="Autocorrelation Function")
lag12 <- acf(res, lag.max=12, plot=FALSE)$acf[13]
bound <- 1.96/sqrt(36)
cat("Autocorrelation at Lag 12:", lag12, "\n")
cat("95% Bound:", bound, "\n")
if(abs(lag12)>bound) {
  cat("Significant yearly cycle detected")
} else {
  cat("No significant yearly cycle detected")
}
```

Output:



R5. Holt-Winters Forecasting

R code:

```
per_train <- ts(per$Sales[1:30],
                start=c(2023,1), frequency=12)
hw_model <- HoltWinters(per_train)
forecast <- predict(hw_model, n.ahead=6,
                    prediction.interval=TRUE,
                    level=0.95)
print(forecast)
actual <- per$Sales[31:36]
predicted <- forecast[, "fit"]
mape <- mean(abs((actual-predicted)/actual))*100
cat("MAPE:", mape, "%\n")
width <- forecast[, "upr"]-forecast[, "lwr"]
```

```
cat("Band Width at h=1:", width[1], "\n")

cat("Band Width at h=6:", width[6], "\n")

plot(per$Month[1:30], per$Sales[1:30],

    type="l", col="blue", lwd=2,

    xlim=range(per$Month),

    ylim=range(c(per$Sales, forecast)),

    xlab="Month", ylab="Sales",

    main="Holt-Winters Forecast")

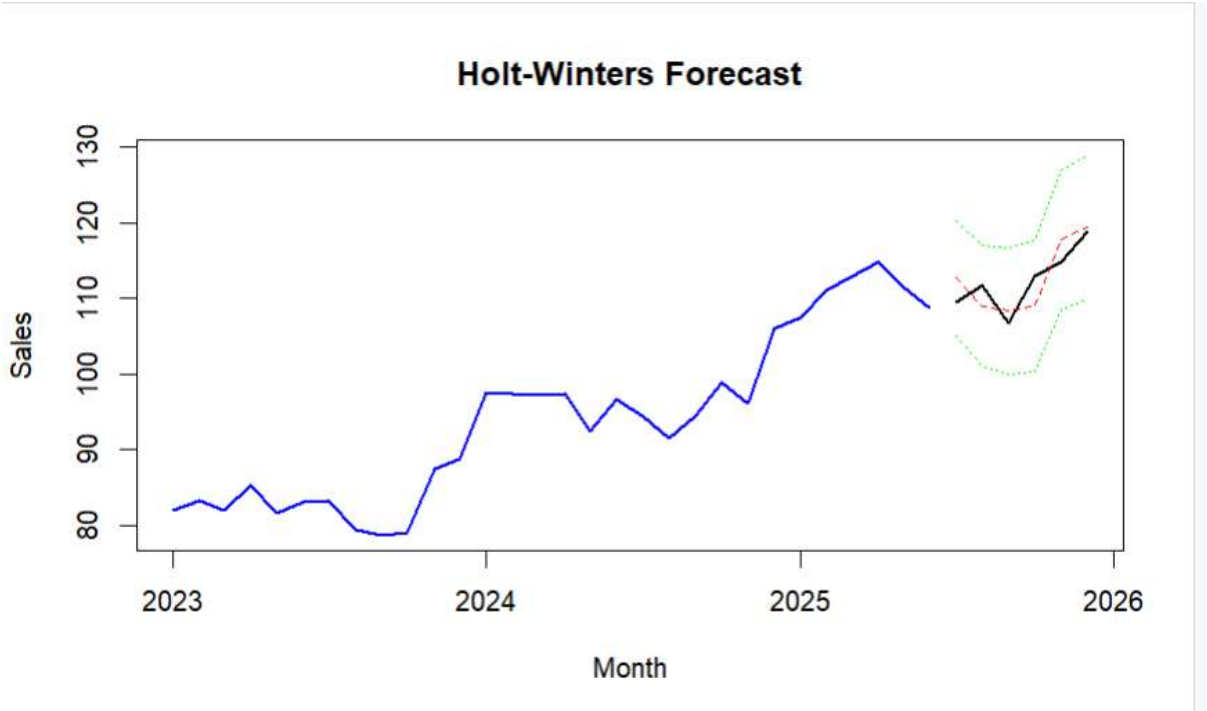
lines(per$Month[31:36], actual, col="black", lwd=2)

lines(per$Month[31:36], predicted, col="red", lty=2)

lines(per$Month[31:36], forecast[,"lwr"], col="green", lty=3)

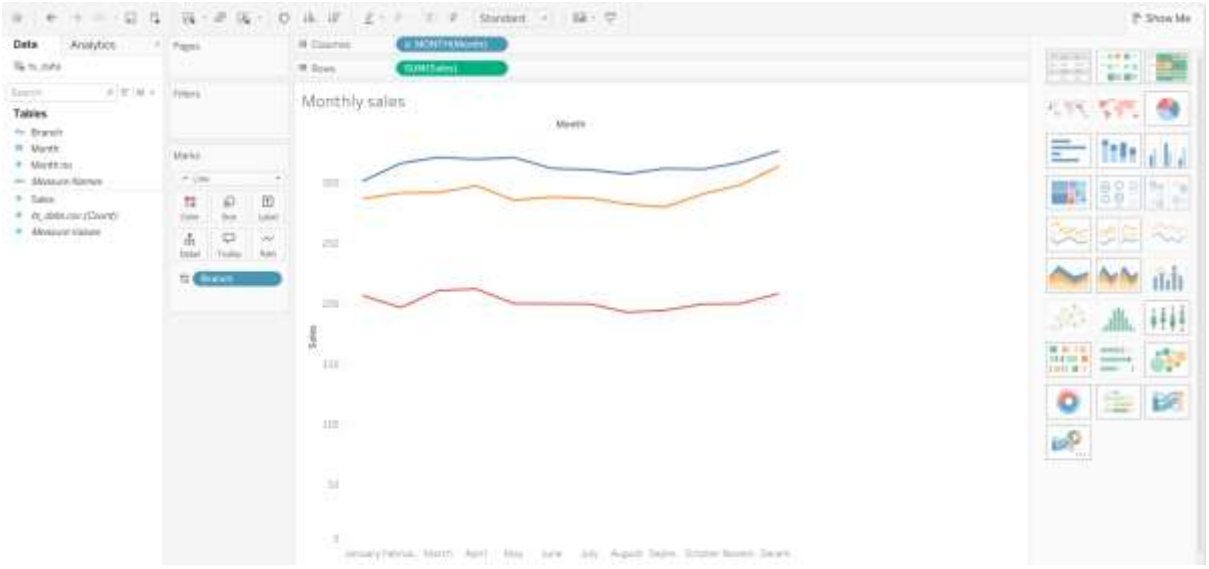
lines(per$Month[31:36], forecast[,"upr"], col="green", lty=3)
```

Output:



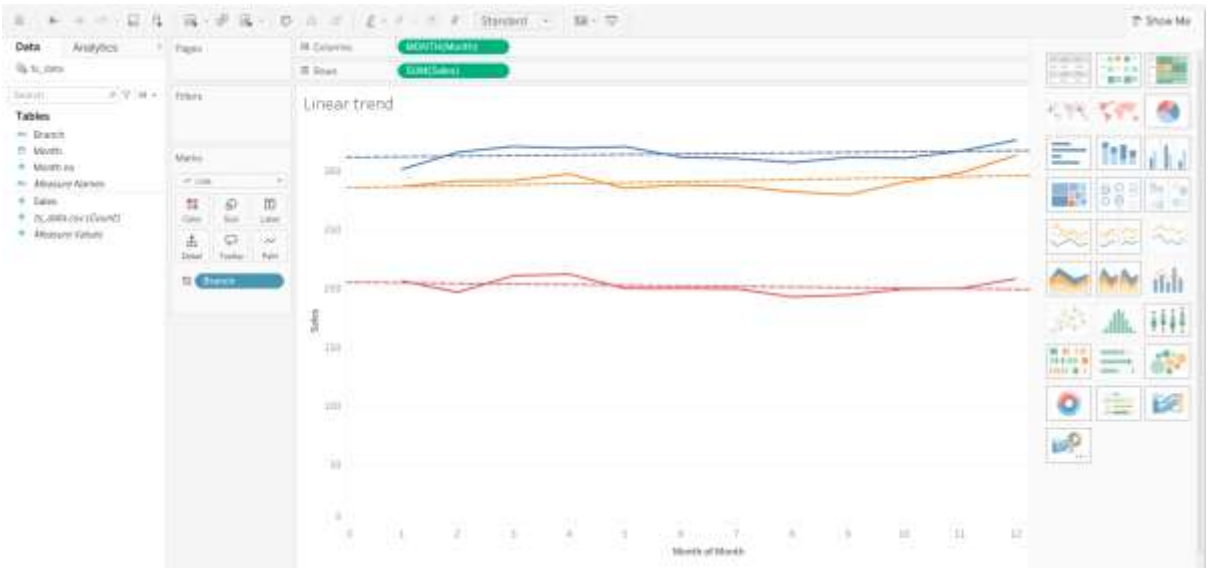
PART 2 – TABLEAU PUBLIC (15 MARKS)

T1 – Monthly Sales by Branch



- Branch with highest sales in December 2025.
- Its December 2025 sales value.

T2 – Linear Trend Lines

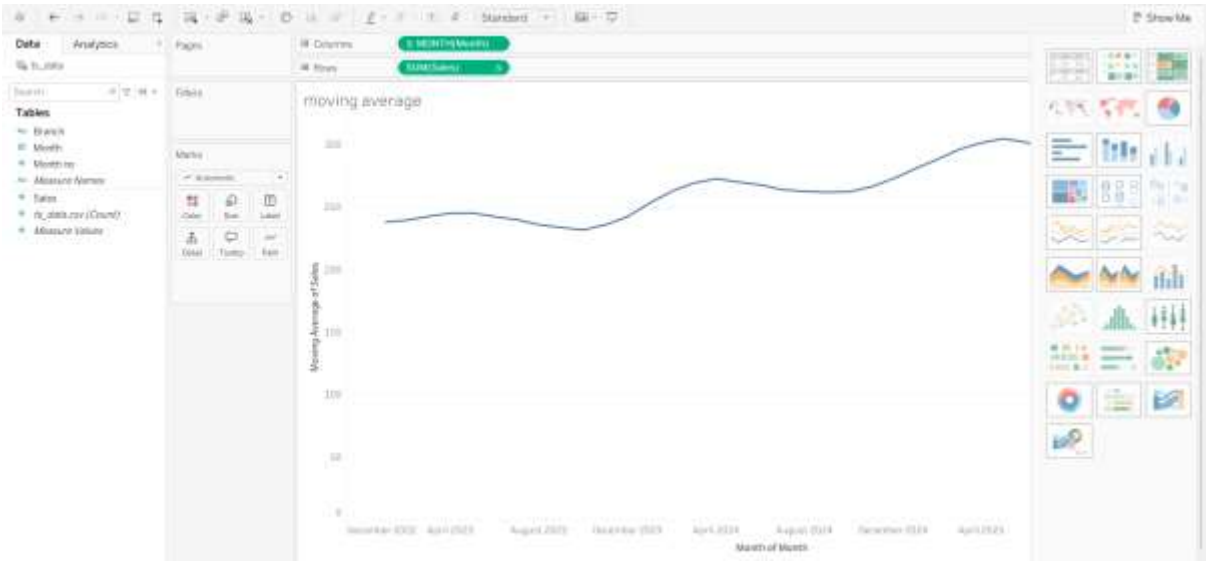


Branch	R-squared
Perambur	0.120721
Velachery	0.0817724
Anna Nagar	0.0593686

- Highest R-squared: Perambur.

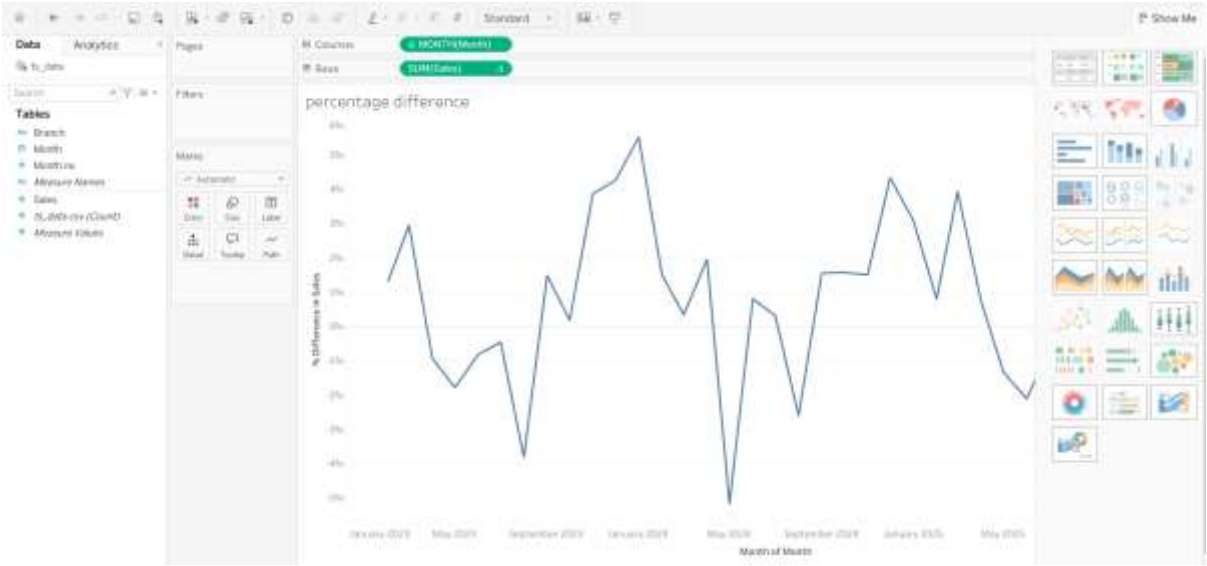
- Lowest R-squared: Anna Nagar.

T3 – Moving Average



Particulars	Value
Tableau moving average at December 2024	272.87

T4 – Percentage Difference



T5 – Differences Between R and Tableau

R Programming

1. R uses programming code for data analysis.

2. It performs statistical calculations and forecasting.
3. It supports customized graphs.
4. It is useful for mathematical analysis.

Tableau

1. Tableau is a data visualization tool.
2. It creates charts using drag-and-drop options.
3. It provides interactive dashboards.
4. It is useful for presenting data visually.